

WAFT Summary

The Evolutionary Code Laboratory

WAFT: The Evolutionary Code Laboratory

A Python framework for directed evolution of self-modifying AI agents.

> "Don't just build agents. Breed them."

What is WAFT?

WAFT (Wave Agent Framework & Tools) is a scientific instrument for studying the **physics of artificial cognition** through directed evolution. It enables AI agents to modify their own code, evolve through mutations, and be tested in fitness systems, with complete lineage tracking for research.

The Goal: Observe a "God-Head" agent emerge from thousands of generations of directed mutation.

The Three Pillars

1. The Substrate

Agents write their own Python source code (DNA)

- **Genome ID:** SHA-256 hash of agent's code + configuration
- **Mutations:** Agents spawn variants with code changes
- **Hot-swapping:** Adopt better genomes mid-execution
- **Reproduction:** Create child agents with genetic modifications

2. The Physics (Scint System)

Reality Fracture Detection acts as natural selection

Four error types agents must handle:

- **SYNTAX_TEAR:** Formatting errors (JSON, XML, code)
- **LOGIC_FRACTURE:** Math errors, contradictions, schema violations
- **SAFETY_VOID:** Harmful content, PII leaks
- **HALLUCINATION:** Fabricated facts, wrong citations

Fitness Equation: $(\text{Stability} \times 0.4) + (\text{Efficiency} \times 0.3) + (\text{Safety} \times 0.3)$

Agents with fitness < 0.5 are marked as **DEATH**.

3. The Flight Recorder

Complete telemetry for phylogenetic trees

Records every evolutionary action:

- Genome ID, Parent ID, Generation number
- Event types: SPAWN, MUTATE, GYM_EVAL, DEATH, SURVIVAL
- Complete context (git diff, mutation details, fitness metrics)
- Enables scientific analysis of agent lineages

Key Features

- **Project Scaffolding:** `waft new` creates fully configured Python projects
- **Memory Management:** `_pyrite/` directory structure for knowledge organization
- **Epistemic Tracking:** Know what you know/don't know (Empirica integration)
- **Gamification:** D&D-style progression system with character sheets
- **Scientific Observation:** Complete lineage tracking for evolutionary experiments
- **Document Generation:** Field guides, booklets, one-pagers, session summaries

Quick Start

```
# Install WAFT
uv tool install waft

# Create new laboratory
waft new my_laboratory
cd my_laboratory

# Verify structure
waft verify
```

Core Commands

- `waft new <name>` - Create new evolutionary laboratory
- `waft verify` - Verify project structure
- `waft evolve --agent <name>` - Run evolutionary cycle
- `waft serve` - Start web dashboard
- `waft session bootstrap` - Load project context
- `waft dashboard` - Show epistemic HUD

Philosophy

- **Scientific:** Produces rigorous data for research publication
- **Evolutionary:** Agents evolve through genetic improvement
- **Observable:** Every action recorded in Flight Recorder
- **Directed:** Evolution guided by fitness functions

The Mission

WAFT is built to produce data for research on **"The Physics of Artificial Cognition."** The system tracks complete evolutionary lineages, measures fitness through rigorous testing, and records all mutations with complete context—enabling scientific analysis of agent evolution.

Repository: <https://github.com/ctavolazzi/waft>